

Alexander Alemi

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URL: alexalemi.com

SCHOLAR: [google scholar](#)

CV: [Curriculum Vitae](#)

WORK: [anthropic](#)

GITHUB: [alexalemi](#)

BLOG: thephysicsvirtuosi.com (0.5M pageviews)

PHYSICS.SE: [alemi](#) (second most popular answer)

EDUCATION & EXPERIENCE

2026-	MEMBER OF TECHNICAL STAFF at Anthropic.
2018-2025	SENIOR RESEARCH SCIENTIST at Google.
2016-2018	SOFTWARE ENGINEER and Machine Learning Researcher at Google.
2015-2016	POSTDOC at Disney Research, Boston.
2015	PHD in Theoretical Physics, Cornell University. Advisor: Jim Sethna
2015	RESEARCH INTERNSHIP for Google.
2014	CONSULTANT for Disney Research.
2014	INTERNSHIP at Disney Research, Boston. Advisor: Jonathan Yedidia
2012	MS in Physics, Cornell University.
2009	BS in Physics, California Institute of Technology, with honors.

EXPERTISE

Information Theory • Machine Learning • Neural Networks • Bayesian Statistics • Numerical Methods • Python • Condensed Matter Physics

PROGRAMMING

Daily:	Python • JAX • (Numpy/Scipy/Matplotlib)
Proficient:	LaTeX • Mathematica • Nim • Julia • Rust • Go • Javascript • Lua • HTML • CSS • C • SQL • Linux • Git • Vim • TensorFlow

RESEARCH

Current:	Information Theory and Deep Learning: Attempting to understand and improve deep networks by understanding the intersection of information theory and deep learning.
Past:	Graph Embeddings • Variational Autoencoders • Variational Information Bottleneck • Image Classification • Classification of the arXiv • Zombie Modelling • Structure of Neural Networks • Smooth Robot Trajectory Optimization • OpenKIM.org Developer • Nonlinear Elasticity of Graphene • Stock Market Structure • Defect Modelling in Atomic Sheets • Elastic Model of Skin Cancer • Nanoelectromechanical Oscillators • Near-field Heat Transfer • Venus' Missing Moon
Ask Me About:	Inverting the Game of Life • Solving Crosswords without Clues • Portal Physics • Effectiveness of Speeding • Sounds of Coins • Racquetball Ping Sound • Battleship • Twitter Compression

SERVICE

- 2024-
2023-2026 MENTOR, Take Stock in Children.
- 2023-2026 PLANNING ADVISORY BOARD, City of Kissimmee, FL. Three-year appointed term reviewing zoning, land development, and comprehensive plan amendments.
- 2022- ACTION EDITOR, Transactions on Machine Learning Research (TMLR).